Chapter 5A Further Aspects of Stratified Sampling

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Effects of errors in the stratum weights

Instead of the true stratum proportions W_h, we have estimates w_h.

- The sample estimate of $\bar{Y}$: $\sum_{h=1}^{L} w_h \bar{y}_h$
- Consequence 1

  The sample estimate is biased. The bias remains constant as the sample size increases.

  $$\sum_{h=1}^{L} (w_h - W_h)\bar{Y}_h$$ independent of n

  The MSE:

$$MSE(\bar{y}_{st}) = V(\bar{y}_{st}) + B^2 = \sum_{h=1}^{L} \frac{w_h^2 S_h^2}{n_h}(1 - f_h) + \left[\sum_{h=1}^{L} (w_h - W_h)\bar{Y}_h\right]^2$$

- Consequence 2

  With the increase of n, the first term will decrease, but the second term remains the same.

  All the gain in precision from stratification will be lost.

- $\sum_{h=1}^{L} w_h \bar{y}_h$ is not an available estimator.
- Accurate estimation of the W_h is particularly important when stratification is highly effective or when the sample size is large.

Effects of deviations from the optimum allocation

For given n, the sample size n_h' in the stratum h should be

$$n_h' = \frac{n(W_h S_h)}{\sum W_h S_h}$$

The resulting minimum variance is

$$V_{\min}(\bar{y}_{st}) = \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2 - \frac{1}{N}\sum_{h=1}^{L} W_h S_h^2$$

If $\hat{n}_h$ is the sample sized used in stratum h,

$$V(\bar{y}_{st}) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{\hat{n}_h} - \frac{1}{N}\sum_{h=1}^{L} W_h S_h^2$$

- The increase in variance caused by the imperfect allocation :

$$V(\bar{y}_{st}) - V_{\min}(\bar{y}_{st}) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{\hat{n}_h} - \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2$$

- According to Neyman allocation: $W_h S_h = \frac{n'_h}{n} \sum_{h=1}^{L} W_h S_h$

$$V(\bar{y}_{st}) - V_{\min}(\bar{y}_{st}) = \sum_{h=1}^{L} \frac{n_h'^2 (\sum W_h S_h)^2}{n^2 \hat{n}_h} - \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2$$

$$= \frac{(\sum_{h=1}^{L} W_h S_h)^2}{n^2}\left(\sum_{h=1}^{L} \frac{n_h'^2}{\hat{n}_h} - n\right) = \frac{(\sum_{h=1}^{L} W_h S_h)^2}{n^2}\left(\sum_{h=1}^{L} \frac{n_h'^2}{\hat{n}_h} - 2n + n\right)$$

$$= \frac{(\sum_{h=1}^{L} W_h S_h)^2}{n^2}\left(\sum_{h=1}^{L} \frac{n_h'^2}{\hat{n}_h} - 2\sum_{h=1}^{L} n'_h + \sum_{h=1}^{L} \hat{n}_h\right) = \frac{(\sum_{h=1}^{L} W_h S_h)^2}{n^2} \sum_{h=1}^{L}\left(\frac{n_h'^2}{\hat{n}_h} - 2n'_h + \hat{n}_h\right)$$

$$= \frac{(\sum_{h=1}^{L} W_h S_h)^2}{n^2} \sum_{h=1}^{L} \frac{(\hat{n}_h - n'_h)^2}{\hat{n}_h}$$

$$V(\bar{y}_{st}) - V_{\min}(\bar{y}_{st}) = \frac{(\sum_{h=1}^{L} W_h S_h)^2}{n^2} \sum_{h=1}^{L} \frac{(\hat{n}_h - n'_h)^2}{\hat{n}_h}$$

- Omit the fpc,

$$V_{\min}(\bar{y}_{st}) = \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2 - \frac{1}{N}\sum_{h=1}^{L} W_h S_h^2 \approx \frac{1}{n}\left(\sum_{h=1}^{L} W_h S_h\right)^2$$

- The proportional increase in variance resulting from deviations from the optimum allocation:

$$\frac{V(\bar{y}_{st}) - V_{\min}(\bar{y}_{st})}{V_{\min}(\bar{y}_{st})} \approx \frac{1}{n}\sum_{h=1}^{L} \frac{(\hat{n}_h - n'_h)^2}{\hat{n}_h}$$

- The optimum can be described as flat!

Effects of deviations from optimum allocation

Stratum	n'_h	$\hat{n}_h$	$\frac{(\hat{n}_h - n'_h)^2}{\hat{n}_h}$
1	200	150	16.7
2	100	120	3.3
3	40	70	12.9
Total	340	340	32.9

- The proportional increase in variance is $32.9/340=9.7\%$.

The problem of allocation with more than one item

- The best allocation for one item will not in general be best for another
- Some compromise must be reached with numerous items

A procedure

Step 1: select the most important items

Step 2: compute the optimum allocation for each item separately, and see to what extent there is disagreement

Step 3: for highly correlated items, taking average among the allocations could be considered



Example

Data given by Jessen (1942) illustrate a farm survey of this kind. The state of Iowa was divided into five geographic regions, each denoted by its major agricultural enterprise. Suppose that these regions are to be used as strata in a survey on dairy farming.

The three items of most interest are (1) the number of cows milked per day, (2) the number of gallons of milk per day, and (3) the total annual cash receipts from dairy products.

The estimated standard deviations s_h within strata, and the optimum Neyman allocations for the individual items are given.

Suppose $n=1000$.

STANDARD DEVIATIONS WITHIN STRATA

Stratum	$W_h = \frac{N_h}{N}$	s_h Cows Milked	s_h Gallons of Milk	s_h Receipts for Dairy Products ($)
Northeast dairy	0.197	4.6	11.7	332
Cash grain	0.191	3.4	9.8	357
Western livestock	0.219	3.3	7.0	246
Southern pasture	0.184	2.8	6.5	173
Eastern livestock	0.208	3.7	9.8	279

SAMPLE SIZES WITHIN STRATA ($n = 1000$)

Stratum	Allocation				
		Optimum for			
	Proportional	Cows	Gallons	Receipts	Average m_h
Northeast dairy	197	254	258	236	250
Cash grain	191	182	209	246	212
Western livestock	219	203	171	194	189
Southern pasture	184	145	134	115	131
Eastern livestock	208	216	228	209	218

EXPECTED VARIANCES OF THE ESTIMATED MEAN

Type of allocation	Cows	Gallons	Receipts
Optimum	0.0127	0.0800	76.9
Compromise	0.0128	0.0802	77.6
Proportional	0.0131	0.0837	80.9

- The compromise allocation gives results almost as precise as if it were possible to use separate optimum allocations for each item.
- Proportional allocation is only slightly less precise than the compromise or the individual optima.
- The precision of the optima and of the compromise is overestimated.
- **The flatness of the optimum**

- Chatterjee's method : an alternative compromise allocation
- To choose the n_h that minimize the average of the proportional increases in variance

$$V_{r_j} = \frac{V(\bar{y}_{st}) - V_{\min}(\bar{y}_{st})}{V_{\min}(\bar{y}_{st})} = \frac{1}{n}\sum_{h=1}^{L}\frac{(n_h - n'_{jh})^2}{n_h}, \qquad j = 1,2,\cdots,k$$

Minimize $\frac{1}{k}\sum_{j=1}^{k} V_{r_j}$ ⇒ $n_h = n\sqrt{\sum_j n'^2_{jh}} \Big/ \sum_h \sqrt{\sum_j n'^2_{jh}}$

- If the individual optima differ only slightly, Chatterjee's n_h vary from the average m_h , at most, one unit in any stratum.

The construction of strata

- What is the best characteristic for the construction of strata?
- How should the boundaries between the strata be determined?
- How many strata should there be?

- For a single item or variable y, the best characteristic is clearly the frequency distribution of y itself.
- The next best : the frequency distribution of some other quantity highly correlated with y.

- Let y_0, y_L be the smallest and largest values of y in the population
- The problem is to find intermediate stratum boundaries $y_1, y_2, \ldots, y_{L-1}$
- A quick approximate method
- $f(y)$: the frequency function of y
- **Rule**: is to form the cumulative of $\sqrt{f(y)}$ and choose the y_h so that they create equal intervals on the cum $\sqrt{f(y)}$ scale.

Example

The data show the frequency distribution of the percentage of bank loads devoted to industrial loans in a population of 13,435 banks.

CALCULATION OF STRATUM BOUNDARIES BY THE CUM $\sqrt{f(y)}$ RULE

Industrial Loans / Total Loans %	$f(y)$	Cum $\sqrt{f(y)}$	Industrial Loans / Total Loans %	$f(y)$	Cum $\sqrt{f(y)}$
0–5	3464	58.9	50–55	126	340.3
5–10	2516	109.1	55–60	107	350.6
10–15	2157	155.5	60–65	82	359.7
15–20	1581	195.3	65–70	50	366.8
20–25	1142	229.1	70–75	39	373.0
25–30	746	256.4	75–80	25	378.0
30–35	512	279.0	80–85	16	382.0
35–40	376	298.4	85–90	19	386.4
40–45	265	314.7	90–95	2	387.8
45–50	207	329.1	95–100	3	389.5

$$389.5 / 5 = 77.9$$

- The division points should be at 77.9,155.8, 233.7, and 311.6 on this scale.

	Stratum				
	1	2	3	4	5
Boundaries	0–5%	5–15%	15–25%	25–45%	45–100%
Interval on cum $\sqrt{f}$	58.9	96.6	73.6	85.6	74.8

- The first two intervals are rather unequal, but cannot be improved on without a finer subdivision of the original classes.
- If the measured variable is x, the cum $\sqrt{f(y)}$ applied to x should give an efficient stratification for y that has a linear regression on x with high correlation.

Number of strata

- At what rate does the variance of $\bar{y}_{st}$ decrease as L is increased?
- The simplest case: $y \sim \text{Uniform}(a, a+d)$

  The population variance: $S_y^2 = d^2/12$
- For a simple random sample of size n: $V(\bar{y}) = d^2/(12n)$
- If L strata of equal sizes are created, $S_{yh}^2 = d^2/12L^2$.

$$\boxed{V(\bar{y}_{st})} = \sum_{h=1}^{L} \frac{1}{n_h} W_h^2 S_{yh}^2 = \frac{1}{n} \sum_{h=1}^{L} W_h \frac{d^2}{12L^2} = \frac{1}{n} \frac{d^2}{12L^2} = \boxed{\frac{V(\bar{y})}{L^2}}$$

The variance decreases inversely
as the square of the number of strata.

- Use some other variable x to construct the strata?

$$V(\bar{y}_{st}) \geq \frac{S_y^2}{n}\left[\frac{\rho^2}{L^2} + (1-\rho^2)\right]$$

where ρ is the correlation between y and x in the unstratified population.

- Unless ρ exceeds 0.95, little reduction in variance is to be expected beyond $L = 6$.
- If an increase in L beyond 6 necessitates any substantial decrease in n in order to keep the cost constant, the increase will seldom be profitable.

$V(\bar{y}_{st})/V(\bar{y})$ AS A FUNCTION OF L FOR THE LINEAR REGRESSION MODEL AND FOR SOME ACTUAL DATA

	Linear Regression Model				Data, Set		
	$\rho =$						
L	0.99	0.95	0.90	0.85	1	2	3
2	0.265	0.323	0.392	0.458	0.197	0.295	0.500
3	0.129	0.198	0.280	0.358	0.108	0.178	0.375
4	0.081	0.154	0.241	0.323	0.075	0.142	0.244
5	0.059	0.134	0.222	0.306	0.065	0.105	0.241
6	0.047	0.123	0.212	0.298	0.050	0.104	0.212
∞	0.020	0.098	0.190	0.277	—	—	—

Stratification after selection of the sample (poststratification)

- The units can be classified into the strata only after the sample data are known.
- One procedure: take a simple random sample of size n and classify the units.
- The stratum sizes N_h may be obtainable fairly accurately from official statistics.
- Assume W_h, N_h are known.

- Estimate the population mean: **not** by the sample mean $\bar{y}$

- But use $$\bar{y}_w = \sum_{h=1}^{L} W_h \bar{y}_h$$

  where $\bar{y}_h$ is the mean of the sample units that fall in stratum h.

- m_h : the number of units in the sample that fall in stratum h.

Note: If one or more of the m_h is zero, two or more strata would have to be combined before making the estimate, and a less precise estimate would be produced.

- Almost as precise as *proportional* stratified sampling, provided that (a) the sample is reasonably large (>20) in every stratum, and (b) the effects of errors in the weights can be ignored.

- For samples in which the m_h are fixed and all m_h exceed zero,

$$V(\bar{y}_W) = \sum_{h=1}^{L} \frac{W_h^2 S_h^2}{m_h} - \frac{1}{N} \sum_{h=1}^{L} W_h S_h^2$$

- Since m_h will vary from sample to sample, we average $V(\bar{y}_W)$

$$E\left[V(\bar{y}_W)\right] \approx \frac{1-f}{n} \sum_{h=1}^{L} W_h S_h^2 + \frac{1}{n^2} \sum_{h=1}^{L} (1-W_h) S_h^2 = V_{prop} + \frac{1}{n^2} \sum_{h=1}^{L} (1-W_h) S_h^2$$

The second term represents the increase in variance that arises because the m_h do not distribute themselves proportionally.

- Variance estimation

$$v(\bar{y}_W) \approx \frac{1-f}{n} \sum_{h=1}^{L} W_h s_h^2 + \frac{1}{n^2} \sum_{h=1}^{L} (1-W_h) s_h^2$$



Example

We are interested in estimating the amount spent in book purchase by the students in a college. A simple random sample of size 200 was selected from total 8000 students. According to some documents in office of educational administration, the proportion of undergraduate students is 70%.

The following table shows the results given by a recent survey. Estimate the average amount per student.

	W_h	n_h	$\bar{y}_h$	s_h
Undergraduate	0.7	120	253.4	231
Graduate	0.3	80	329.4	367
Totals	1	200	283.8	294.57

The estimated average expenditure per student:

$$\bar{y}_w = \sum_{h=1}^{2} W_h \bar{y}_h = 0.7 \times 253.4 + 0.3 \times 329.4 = 276.2$$

The variance:

$$v(\bar{y}_w) \approx \frac{1-f}{n} \sum_{h=1}^{2} W_h s_h^2 + \frac{1}{n^2} \sum_{h=1}^{2} (1-W_h) s_h^2$$

$$= \frac{1-0.025}{200} (0.7 \times 231^2 + 0.3 \times 367^2) + \frac{1}{200} (0.3 \times 231^2 + 0.7 \times 367^2)$$

$$= 381.83$$

Standard error: $se(\bar{y}_w) = 19.54$

If we use simple estimate, then the variance is

$$v(\bar{y}) \approx \frac{1-f}{n} s^2 = \frac{1-0.025}{200} \times 294.57^2 \approx 423.01$$

Standard error: $se(\bar{y}) = 20.57$

Note: This method can also be applied to a sample that is already stratified by another factor.